Chapter 2: Vector Spaces

Worked Answer Key Part 1 of 5

Solutions correspond to Exercises 1–20 in the companion set. All vector spaces are over the real numbers.

A. Vector spaces and subspaces

1. S_1 is a subspace: the defining expression is linear and homogeneous, so it remains zero under any linear combination. Solving for x gives

$$(x, y, z)^T = y(-2, 1, 0)^T + z(1, 0, 1)^T.$$

These two vectors span S_1.

S_2 is not a subspace because it does not contain $\mathbf{0}$.

S_3 is not a subspace: $(1, 0, 0)^T$ and $(0, 1, 0)^T$ belong to it, but their sum $(1, 1, 0)^T$ does not.

2. A vector $t(1, 0, 1)^T$ lies in P only when its third component t is zero. Hence $P \cap L = \{\mathbf{0}\}$. The union is not a subspace: $(0, 1, 0)^T \in P$ and $(1, 0, 1)^T \in L$, but their sum $(1, 1, 1)^T$ lies in neither.

Since $L_0 \subseteq P$, we have $P \cap L_0 = L_0$ and $P \cup L_0 = P$. Both are subspaces.

3. A column combination has the form

$$D\begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} s \\ t \\ s + t \end{pmatrix}.$$

Thus $C(D) = \{(u, v, w)^T : w = u + v\}$, a plane through the origin. The vector $\mathbf{b}$ belongs to it because $1 = 2 + (-1)$, whereas $\mathbf{c}$ does not. The unique solution of $D\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = (2, -1)^T$.

4. First, $A\mathbf{0} = \mathbf{0}$, so $N(A)$ is nonempty. If $A\mathbf{u} = A\mathbf{v} = \mathbf{0}$, then

$$A(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha A\mathbf{u} + \beta A\mathbf{v} = \mathbf{0}.$$

Thus $N(A)$ is closed under addition and scalar multiplication, and the other vector-space axioms are inherited from $\mathbb{R}^n$.

For $\mathbf{b} \neq \mathbf{0}$, the zero vector is not a solution, so the solution set is not a subspace. Containing zero is not sufficient: S_3 in Exercise 1 contains zero but is not closed under addition.

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B. Echelon form and nullspaces

5. Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$A \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

Then apply $R_3 \leftarrow R_3 - R_2$ and $R_1 \leftarrow R_1 - R_2$:

$$R = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The rank is 2; columns 1 and 3 are pivot columns. The basic variables are x_1, x_3, and the free variables are x_2, x_4.

6. The reduced equations are $x_1 + 2x_2 + x_4 = 0$ and $x_3 + 2x_4 = 0$. Setting $x_2 = s$ and $x_4 = t$ gives

$$\mathbf{x} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -2 \\ 1 \end{pmatrix}, \qquad N = \begin{pmatrix} -2 & -1 \\ 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{pmatrix}.$$

The columns are independent, as seen from their free-variable components. Direct multiplication gives $AN = 0_{3\times 2}$. Therefore $\dim N(A) = 4 - 2 = 2$: a two-dimensional plane through the origin in $\mathbb{R}^4$.

7. The column order becomes $(1, 3, 2, 4)$, so

$$Q = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad F = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} -F \\ I_2 \end{pmatrix} = \begin{pmatrix} -2 & -1 \\ 0 & -2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Multiplication by Q interchanges its second and third rows and gives the matrix N in Exercise 6. Without Q, the entries refer to (x_1, x_3, x_2, x_4) rather than (x_1, x_2, x_3, x_4).

8. Use $R_3 \leftarrow R_3 - R_1 - R_2$ and $R_4 \leftarrow R_4 - 2R_1 + R_2$ to obtain

$$\operatorname{rref}(B) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\operatorname{rank}(B) = 2$, and all nullspace vectors are $t(-2, -1, 1)^T$. This vector is a basis for $N(B)$, whose dimension is 1. Although $m = 4 > 3 = n$, the equations are dependent and $r = 2 < n$, leaving one free variable.

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C. Consistency and complete solution sets

9. Applying the operations from Exercise 5 to the right-hand side gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 3b_1 - b_2 \\ 0 & 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_1 - b_2 + b_3 \end{array}\right).$$

Hence consistency is equivalent to $b_1 - b_2 + b_3 = 0$, and

$$C(A) = \{(b_1, b_2, b_3)^T : b_1 - b_2 + b_3 = 0\}.$$

The first proposed right-hand side satisfies $2 - 5 + 3 = 0$ and is consistent. The second gives $2 - 5 + 4 = 1$, so it is inconsistent.

10. The reduced equations are $x_1 + 2x_2 + x_4 = 1$ and $x_3 + 2x_4 = 1$. Setting $x_2 = x_4 = 0$ gives $\mathbf{x}_p = (1, 0, 1, 0)^T$. Indeed, $A\mathbf{x}_p = A_{*1} + A_{*3} = (2, 5, 3)^T$.

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -2 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{R}.$$

This is a two-dimensional affine plane in $\mathbb{R}^4$, parallel to $N(A)$. It does not pass through the origin because the right-hand side is nonzero.

11.

Case	Solvable for every $\mathbf{b}$?	Possible outcomes	$\dim N(A)$	$\dim N(A^T)$
(a)	No	Zero or one	0	1
(b)	Yes	Infinitely many	2	0
(c)	No	Zero or infinitely many	3	2

Full row rank ($r = m$) guarantees consistency for every right-hand side; only (b) has this property. Full column rank ($r = n$) gives uniqueness whenever consistent; only (a) has this property. The two nullspace dimensions are $n - r$ and $m - r$, respectively.

12. Subtracting the first and second equations from the third gives $(a - 1)z = \beta - 3$.

- If $a \neq 1$, set $q = (\beta - 3)/(a - 1)$. The unique solution is $(x, y, z) = (q - 1, 2 - q, q)$.
- If $a = 1$ and $\beta = 3$, all solutions are $(x, y, z) = (t - 1, 2 - t, t)$, $t \in \mathbb{R}$.
- If $a = 1$ and $\beta \neq 3$, the system is inconsistent.

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D. Independence, spanning sets, and bases

13. Put the vectors into columns and subtract rows 1 and 2 from row 3:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & a \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & a-2 \end{pmatrix}.$$

There are three pivots exactly when $a \neq 2$, so the vectors are independent for those values. At $a = 2$, $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$. Their span is the plane $z = x + y$, with basis $\{\mathbf{v}_1, \mathbf{v}_2\}$.

14. We have $\mathbf{w} = \mathbf{u} + \mathbf{v}$ and $\mathbf{q} = 2\mathbf{u} - \mathbf{v}$. Since $\mathbf{u}, \mathbf{v}$ are independent, the span has dimension 2 and one basis is $\{\mathbf{u}, \mathbf{v}\}$. A second is $\{\mathbf{w}, \mathbf{q}\}$: if $c\mathbf{w} + d\mathbf{q} = 0$, then $c + 2d = 0$ and $c - d = 0$, forcing $c = d = 0$.

$$(3, -1, 2)^T = 3\mathbf{u} - \mathbf{v} = \tfrac{1}{3}\mathbf{w} + \tfrac{4}{3}\mathbf{q}.$$

15. At $a = 3$, the final pivot in Exercise 13 is 1, so the three columns are independent and span $\mathbb{R}^3$. For coefficients c_1, c_2, c_3, the target vector gives

$$c_1 + c_3 = 2, \quad c_2 + c_3 = 1, \quad c_1 + c_2 + 3c_3 = 4.$$

The solution is $(c_1, c_2, c_3) = (1, 0, 1)$, so $(2, 1, 4)^T = \mathbf{v}_1 + \mathbf{v}_3$. Uniqueness follows because the difference of two coefficient vectors would be a nullspace vector of a full-rank matrix, hence zero.

16. (a) True. A 3×4 matrix has rank at most 3, so its nullspace has dimension at least 1. Thus its columns have a nontrivial dependence relation.

(b) False. For $D = (1, 1)^T$, the operation $R_2 \leftarrow R_2 - R_1$ produces $(1, 0)^T$. The two column spaces are distinct lines.

(c) False. For D in (b), $m = 2$, $n = 1$, $r = 1$, but $\dim N(D) = 0$, not $m - r = 1$. The correct formula is $n - r$.

(d) True. A square matrix formed from these columns has n pivots. There are no zero rows in its rref, so every right-hand side is a column combination. The vectors are independent and span $\mathbb{R}^n$.

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E. The four fundamental subspaces

17. Using the reduction and special solutions already obtained:

Space	A basis	Dimension	Ambient space
$C(A)$	$\{(1,2,1)^T, (1,3,2)^T\}$	2	$\mathbb{R}^3$
$N(A)$	$\{(-2,1,0,0)^T, (-1,0,-2,1)^T\}$	2	$\mathbb{R}^4$
$C(A^T)$	$\{(1,2,0,1)^T, (0,0,1,2)^T\}$	2	$\mathbb{R}^4$
$N(A^T)$	$\{(1,-1,1)^T\}$	1	$\mathbb{R}^3$

The column and row spaces have dimension $r = 2$. The nullspace dimensions are $n - r = 2$ and $m - r = 1$. The left-nullspace vector satisfies $A^T(1,-1,1)^T = 0$.

18. Applying $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$, $R_3 \leftarrow R_3 - R_2$, and $R_1 \leftarrow R_1 - R_2$ gives

$$[R \mid G] = \left(\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right).$$

Thus $G = \begin{pmatrix} 3 & -1 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$. Multiplication gives the rref R in Exercise 5. Row 3 of G is $(1,-1,1)$ and produces the zero row of R. Therefore $(1,-1,1)^T$ spans $N(A^T)$, whose dimension is 1.

19. Bases are

$$\begin{aligned} C(B): &\quad (1,0,1,2)^T,\ (0,1,1,-1)^T, \\ N(B): &\quad (-2,-1,1)^T, \\ C(B^T): &\quad (1,0,2)^T,\ (0,1,1)^T, \\ N(B^T): &\quad (-1,-1,1,0)^T,\ (-2,1,0,1)^T. \end{aligned}$$

Their dimensions are $2, 1, 2, 2$, respectively. The left-nullspace conditions are $b_3 = b_1 + b_2$ and $b_4 = 2b_1 - b_2$. For $(1,2,3,0)^T$, both hold. A particular solution is $(1,2,0)^T$, so all solutions are

$$\mathbf{x} = (1,2,0)^T + t(-2,-1,1)^T, \qquad t \in \mathbb{R}.$$

20. If $A\mathbf{x} = \mathbf{b}$ and $A^T\mathbf{y} = 0$, then

$$\mathbf{y}^T\mathbf{b} = \mathbf{y}^T A\mathbf{x} = (A^T\mathbf{y})^T\mathbf{x} = 0.$$

For sufficiency, let H be an invertible row-operation matrix with $HA = R$ in echelon form. Every zero row of R is produced by a row $\mathbf{h}_i^T$ of H satisfying $\mathbf{h}_i^T A = 0$. The assumed conditions force each corresponding entry $\mathbf{h}_i^T\mathbf{b}$ to vanish. Thus $[R \mid H\mathbf{b}]$ has no contradictory row and is consistent.

For the specified B and right-hand side, $b_3 = b_1 + b_2$ holds, but $b_4 = 1 \neq 2b_1 - b_2 = 0$. Equivalently, $(-2,1,0,1)\mathbf{b} = 1$. There is no solution.